



ADAM NABEEL

Curriculum vitae

PERSONAL INFORMATION	S/o Dr Majeed Mohamed, SA-08, NAL Campus, CSIR-National Aerospace Laboratories, linkedin.com/in/adamnabeel/	Mobile: +91 9731763225 adamnabeel2004@outlook.com DoB: 25/05/2004 Married: Single
PROFILE	Electronics and Communication Engineering graduate with hands-on experience in GPS-denied navigation, Artificial Intelligence (AI) modeling, and multi-sensor data fusion at the National Aerospace Laboratories (NAL), Bangalore. Proficient in MATLAB/Simulink and Python for avionics, guidance, navigation, and control (GNC) system development, with expertise in machine learning, state estimation, sensor integration, and autonomous aerospace system modeling. In addition, has expertise in Advanced Driver Assistance Systems (ADAS) and Vehicle-to-Vehicle (V2V) communication, supporting intelligent mobility and connected vehicle applications. Demonstrated strong technical capability in the design and implementation of Kalman filter-based estimation algorithms, adaptive Unscented Kalman Filtering, and advanced sensor fusion techniques for autonomous flight vehicles. Experienced in developing innovative navigation and control solutions that enhance accuracy, robustness, and reliability in aerospace applications. Proven ability to apply theoretical concepts to real-world avionics and autonomous systems, delivering effective solutions for advanced aerospace technologies.	
EDUCATION	Visvesvaraya Technological University (VTU), Bangalore B.E., Electronics and Communication Engineering, CGPA 8.1(10.0-Scale) • June-2026 at CMR Institute of Technology, Bengaluru Kendriya Vidyalaya NAL Campus, Bangalore 12th Grade, Maths and computer science, 2022 73% Marks 10th Grade, CBSE, 2020 86% Marks	
TECHNICAL SKILLS	• Programming Languages: MATLAB, Python and Embedded C • Software & Tools: MATLAB/Simulink, Keil μVision, NS-3, Solid Edge, Cadence • Embedded Systems: Arduino, Raspberry Pi, Microcontroller Programming, Real-Time Systems • Domain Skills: State & Parameter Estimation, Sensor Fusion, GPS-Denied Navigation, V2V Communication, Vehicle & UAV Dynamics, ADAS, Neural Network Modelling	
PROJECTS HANDLED	1. Aircraft Avionics System Architecture Simulation Developed and validated comprehensive aircraft avionics system models in MATLAB/Simulink, encompassing altitude regulation, navigation, guidance, and flight path control functions. Performed system-level architecture design, dynamic modelling, simulation, and performance analysis to evaluate operational effectiveness, control accuracy, and integration of critical avionics subsystems. 2. Li-Fi Enabled Radar-Based Collision Prevention System Designed, developed, and validated an ADAS prototype integrating radar and ultrasonic sensors for real-time obstacle detection, collision avoidance, and automatic emergency braking. Implemented embedded control algorithms using Arduino and Embedded C,	

conducted system integration, testing, and performance evaluation to enhance vehicle safety and situational awareness.

3. Multi-Source Vibration Modelling of Elevator Systems for Fault Prediction

Developed a three-degree-of-freedom (3-DOF) dynamic system with multiple excitation sources and acquired vibration response data under varying operating conditions. Extracted frequency-domain features using FFT techniques and generated a comprehensive dataset to support future neural network-based fault detection, classification, and predictive maintenance studies.

INTERNSHIPS

1. **Research Intern at National Institute of Technology, Karnataka March-April 2025.**
 - Developed and evaluated DSRC-based V2V communication algorithms for real-time collision avoidance and vehicular safety applications.
 - Developed and executed network simulation models in NS-3 and performed quantitative performance analysis and validation using MATLAB.
2. **Research Assistant at CSIR-NAL, Bangalore June-July 2025.**
 - Developed vehicle state prediction models for accurate position estimation in V2V communication systems, enhancing situational awareness and collision avoidance performance using MATLAB and Python.
 - Implemented state estimation algorithms incorporating neural network-based sensor models for accurate vehicle position tracking and enhanced localization performance.
3. **Research Assistant at CSIR-NAL, Bangalore Feb-June 2026.**
 - Designed and developed GPS-denied navigation algorithms for autonomous systems, including UAVs, leveraging multi-sensor fusion, state estimation, and AI-based techniques for robust localization and navigation in challenging environments.
 - Developing Unscented Kalman Filter (UKF)-based sensor fusion and nonlinear state estimation algorithms for robust positioning, tracking, and real-time navigation in autonomous systems.

LANGUAGES

- **English** Fluent; **Hindi** Fluent; **Malayalam** Fluent; **Tamil** Fluent; **Kannada** Beginner

CONFERENCE PAPER

- Presented**
- **Adam Nabeel**, G. Selvaraj, M. Mohamed, and M. Pappa, "Machine Learning-Enhanced EKF for Vehicle Localization under Sparse Observations," in Proc. International Conference on Advanced Computing Systems and Intelligent Communication Engineering (ICACSICE 2026)
 - Ergonomic Paper Clip Design, Patent, 2025

PUBLICATIONS:

- **Adam Nabeel**, G. Selvaraj, M. Mohamed, and M. Pappa, "Machine Learning-Enhanced EKF for Vehicle Localization under Sparse Observations," in Proc. **5th International Conference on AI-ML-Systems** (AIMLSystems), 2025,.
- **Adam Nabeel**, "High-Precision Vehicle Localization Using Li-Fi Supported V2V Communication and Sensor Fusion," TIJER – International Research Journal, 2026.
- Ergonomic Paper Clip Design, Patent, 2025
- **Adam Nabeel**, M. Mohamed "UKF and Neural Modelling based GPS denied Navigation for UAV" Submitted for Conference Proceedings.
- **Adam Nabeel**, M. Mohamed "GPS denied Navigation Using Adaptive non-linear filter and Neural Network" in Proc. Journal of Guidance, Control, and Dynamics (JGCD)-2026 (Submitted for Review)

ANY ADDITIONAL INFORMATION

- Served as Class Representative for 8 semesters (2022–2026), facilitating communication between students and faculty.
- Silver Medalist in a Regional-Level Football Tournament.
- Volunteered and hosted events at CULTURA, demonstrating leadership and event coordination skills.
- Represented CMRIT in the VTU State-Level Athletic Meet.

28-06-2026

Adam Nabeel

DEVELOPED GPS DENIED NAVIGATION ALGORITHM FOR UAV

My Original Contributions: (i) UAV data filtering and preprocessing; (ii) problem formulation with the guidance of the supervisor; (iii) integration of the sensor fusion algorithm; (iv) estimation of the dynamic system states and parameters using the Unscented Kalman Filter (UKF); (v) data analysis and neural network training with generation of pseudo GPS data; and (vi) verification and validation of the proposed GPS-denied navigation algorithm.

Description: The proposed system is designed to enable accurate UAV navigation even in GPS-denied environments where traditional GPS-based positioning is not available. The framework combines an Inertial Navigation System (INS), an Unscented Kalman Filter (UKF), and a Neural Network-based error correction module. Initially, the INS uses sensor data from accelerometers and gyroscopes to estimate position, velocity, and orientation. However, since INS errors accumulate over time, the UKF is used to fuse INS data with GPS measurements (when available) to provide accurate and corrected estimates. These GPS-aided outputs act as a reference for understanding system behavior and error patterns.

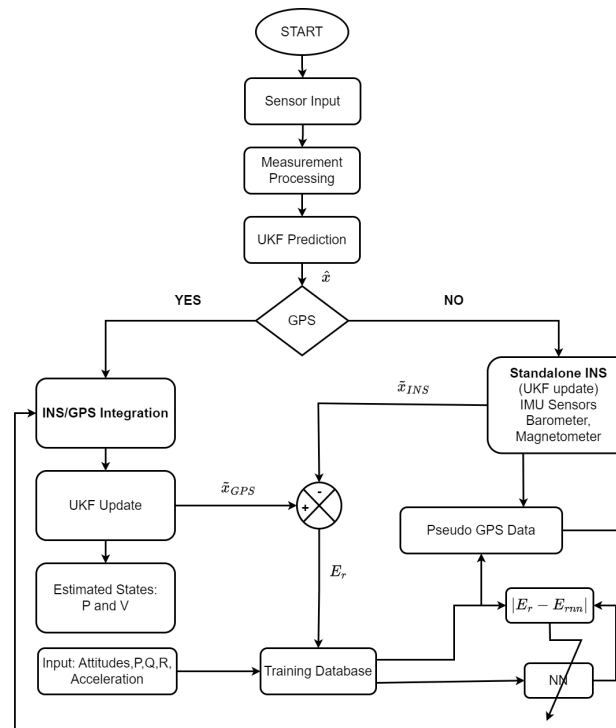


Figure 1: Block Diagram of GPS Denied Navigation Algorithm

During GPS outages, the system operates in standalone INS mode, where errors begin to grow due to the lack of external corrections. To address this, a neural network is trained using the difference between GPS-based and INS-based estimates to learn how errors evolve. Once trained, it predicts these errors in real-time and generates pseudo-GPS signals, which are fed back into the UKF for correction. This approach effectively reduces drift and maintains navigation accuracy. Additional steps such as sensor noise filtering and barometric altitude integration further improve system stability, making the overall framework robust and reliable in challenging environments.

REFERENCES

- Dr.Pappa M, ,
 • HoD, ECE, CMR Institute of Technology, Karnataka. Phone: +91 98865 39880
 • Internal guide at CMRIT. E-mail: hod.ece@cmrit.ac.in
- Dr.Gopinath S, ,
 • HoD, ADAS, UST, Karnataka, India. Phone: +91 98451 68026
 • External guide for vehicle localization. E-mail: selgopi@gmail.com
- Dr.Majeed M,
 • Scientist F, National Aerospace Laboratories, Bangalore. Phone: +91 97317 22655
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